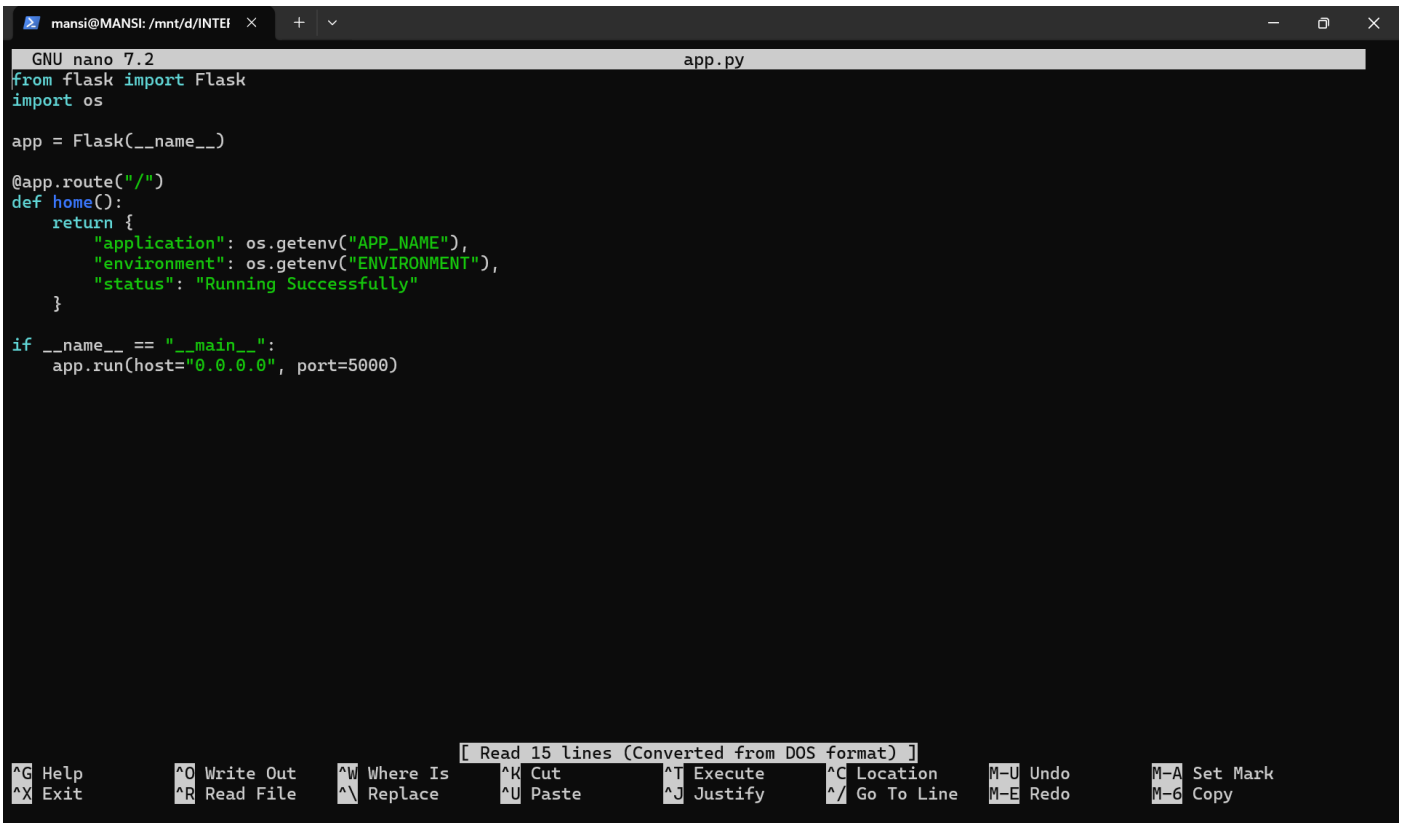


DEVOPS DEPLOYMENT WORKFLOW



```
manshi@MANSI: /mnt/d/INTEI x
GNU nano 7.2 app.py
from flask import Flask
import os

app = Flask(__name__)

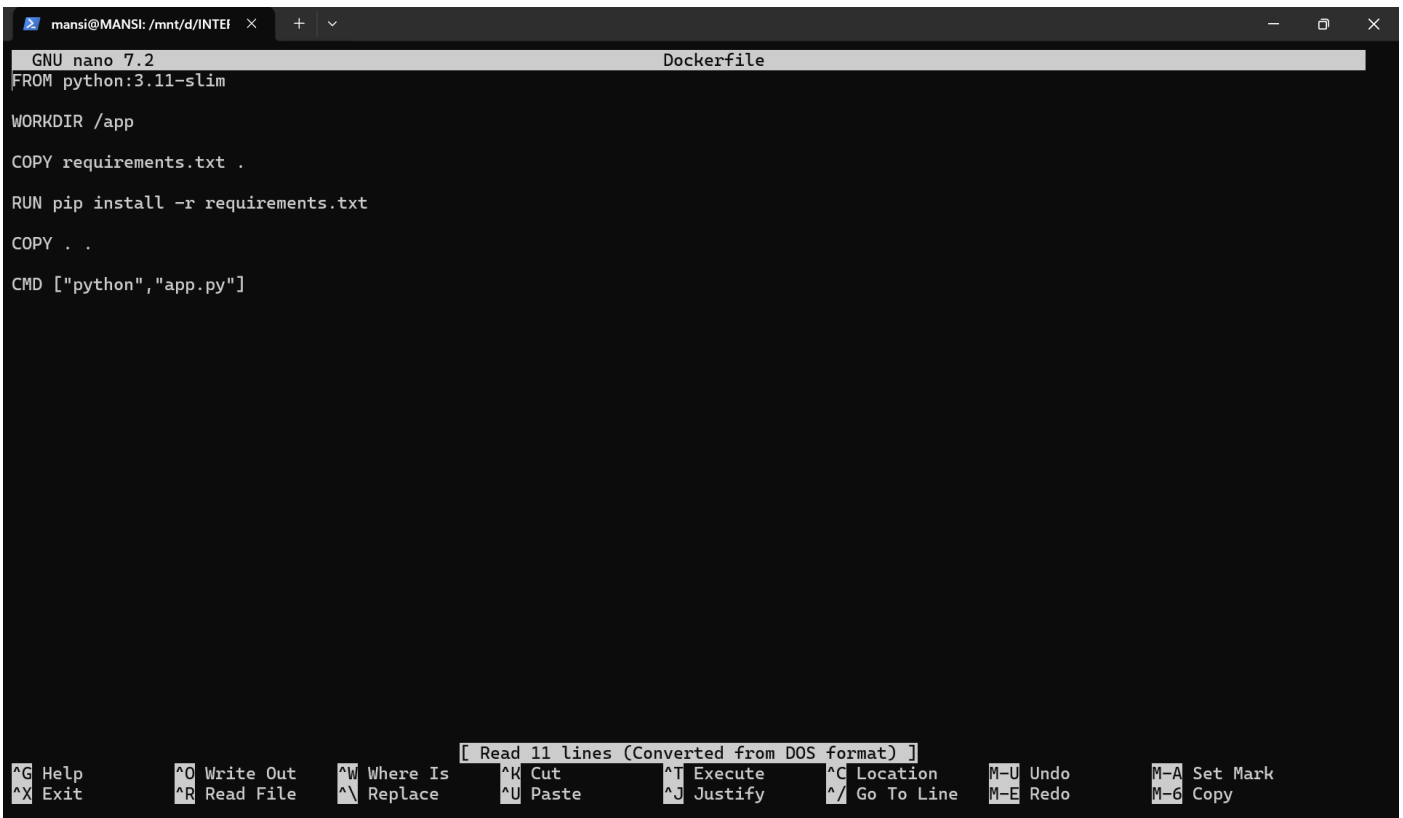
@app.route("/")
def home():
    return {
        "application": os.getenv("APP_NAME"),
        "environment": os.getenv("ENVIRONMENT"),
        "status": "Running Successfully"
    }

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

[Read 15 lines (Converted from DOS format)]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^_ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-G Copy

```
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ nano app.py
manshi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ cat requirements.txt
Flask
```



```
manshi@MANSI: /mnt/d/INTEI x
GNU nano 7.2 Dockerfile
FROM python:3.11-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install -r requirements.txt

COPY . .

CMD ["python", "app.py"]
```

[Read 11 lines (Converted from DOS format)]

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^_ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-G Copy

```
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ nano .env
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ cat .env
APP_NAME=DevOps-Day05
ENVIRONMENT=Production
```

```
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ docker build -t day05-app .
[+] Building 187.4s (10/10) FINISHED
=> [internal] load build definition from Dockerfile                                docker:default 0.1s
=> => transferring dockerfile: 181B                                              0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim              0.1s
=> [internal] load .dockerignore                                                 0.1s
=> => transferring context: 2B                                                  0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:ae52c5bef62a6bdd42cd1e8dffe86b9cd284bde9427da79839de7a4b983e7ca 0.1s
=> => resolve docker.io/library/python:3.11-slim@sha256:ae52c5bef62a6bdd42cd1e8dffe86b9cd284bde9427da79839de7a4b983e7ca 0.1s
=> [internal] load build context                                                0.1s
=> => transferring context: 669B                                                0.0s
=> CACHED [2/5] WORKDIR /app                                                    0.0s
=> [3/5] COPY requirements.txt .                                                0.1s
=> [4/5] RUN pip install -r requirements.txt                                    185.7s
=> [5/5] COPY . .                                                                0.0s
=> exporting to image                                                            1.0s
=> => exporting layers                                                            0.7s
=> => exporting manifest sha256:77d6f30314b4156f0a467c638114424ae95566710a287746f5ef409413b5d2c4 0.0s
=> => exporting config sha256:9ba776841d065057b392b13bfcce726c4ff3bfc05641a5bc6efc81ee5027578f 0.0s
=> => exporting attestation manifest sha256:7721b3d27d9a962e2fb2e4e860c7f5ffe9d20f7b7052c3455a2c1e0290858c62 0.0s
=> => exporting manifest list sha256:932fd0837536c7187070c9d61f4f01f646e43a93e4f1911a4d8aec0f3eda23bb 0.0s
=> => naming to docker.io/library/day05-app:latest                             0.0s
=> => unpacking to docker.io/library/day05-app:latest                           0.2s
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
alpine:latest	a2d49ea686c2	13.1MB	3.95MB	
day03-web:latest	137cb622fb9d	212MB	52MB	
day04-web:latest	bddb035bf67d	212MB	52MB	U
day05-app:latest	932fd0837536	212MB	52MB	
hello-world:latest	940c619fbd41	20.4kB	4.02kB	
my-busybox-app:latest	f628a2465ebd	6.81MB	2.23MB	
my-python-app:latest	3874a0b0427e	186MB	45.4MB	
nginx:alpine	8b1e78743a03	93.6MB	27MB	U
python:3.11-slim	ae52c5bef62a	188MB	47.7MB	

```
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ docker run -d -p 5000:5000 --env-file .env day05-app
de6e506017c2ecd4622ec1f54161580f9274c05feac14178209be0b570ddb46
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS                               NAMES
de6e506017c2  day05-app "python app.py"         11 seconds ago  Up 11 seconds  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp  funny_ride
```

```
localhost:5000
localhost:5000
Pretty-print
{
  "application": "DevOps-Day05",
  "environment": "Production",
  "status": "Running Successfully"
}
```

```
mansi@MANSI: /mnt/d/INTEI x + v
GNU nano 7.2 docker-compose.yml *
services:
  web:
    build: .
    container_name: day05-web

    ports:
      - "5000:5000"

    env_file:
      - .env

    restart: always|

[ Cancelled ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_/ Go To Line M-E Redo      M-G Copy
```

```
mansi@MANSI: /mnt/d/INTERNSHIP/DevOps/Day05$ nano docker-compose.yml
mansi@MANSI: /mnt/d/INTERNSHIP/DevOps/Day05$ docker compose up -d
[+] up 1/1
Container day05-web Started
mansi@MANSI: /mnt/d/INTERNSHIP/DevOps/Day05$ docker compose ps
NAME          IMAGE          COMMAND                  SERVICE    CREATED        STATUS          PORTS
day05-web     day05-web     "python app.py"         web        3 minutes ago  Up 13 seconds  5000/tcp
```

```
mansi@MANSI: /mnt/d/INTEI x + v
GNU nano 7.2 deploy.yml *
name: Day05 Deployment Pipeline

on:
  push:
    branches:
      - main

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout Repository
        uses: actions/checkout@v4

      - name: Verify Files
        run: ls -la

      - name: Docker Build Check
        run: docker --version|
```

```

mansi@MANSI:/mnt/d/INTERNSHIP/DevOps$ git add .
git commit -m "Added Day05 DevOps Task"
git push origin main
[main d6a43ae] Added Day05 DevOps Task
 7 files changed, 88 insertions(+)
 create mode 100644 .github/workflows/deploy.yml
 create mode 100644 Day05/.env
 create mode 100644 Day05/.github/workflows/deploy.yml
 create mode 100644 Day05/Dockerfile
 create mode 100644 Day05/app.py
 create mode 100644 Day05/docker-compose.yml
 create mode 100644 Day05/requirements.txt
Username for 'https://github.com': Radhika-08
Password for 'https://Radhika-08@github.com':
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 8 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (13/13), 1.53 KiB | 39.00 KiB/s, done.
Total 13 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Radhika-08/DevOps.git
 4322e4b..d6a43ae  main -> main

```

The screenshot displays the GitHub Actions interface for the 'Radhika-08 / DevOps' repository. The 'Actions' tab is selected, showing a list of workflow runs. The left sidebar contains navigation links for 'All workflows', 'Day05 Deployment Pipeline', 'DevOps Day04 CI', and various management tools like 'Caches', 'Attestations', 'Runners', 'Usage metrics', and 'Performance metrics'. The main area, titled 'All workflows', shows 4 workflow runs. Each run entry includes a green status icon, the workflow name, the commit hash, the actor, the branch, and the completion time.

Workflow	Event	Status	Branch	Actor	Time
Added Day05 DevOps Task	Day05 Deployment Pipeline #1: Commit d6a43ae pushed by Radhika-08	Success	main	Radhika-08	now 16s
Added Day05 DevOps Task	DevOps Day04 CI #5: Commit d6a43ae pushed by Radhika-08	Success	main	Radhika-08	now 11s
Day04 Screenshots	DevOps Day04 CI #4: Commit 4322e4b pushed by Radhika-08	Success	main	Radhika-08	Today at 3:51 AM 7s
Added Day04 DevOps Task	DevOps Day04 CI #3: Commit c7f9730 pushed by Radhika-08	Success	main	Radhika-08	Today at 3:38 AM 7s

MONITORING, AUTOMATION & CI/CD PIPELINE

```
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS          NAMES
d4bc35ebe5af   day05-web     "python app.py"         12 minutes ago Up 8 minutes   5000/tcp       day05-web
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps$ docker logs day05-web
* Serving Flask app 'app'
* Debug mode: off
* Tip: There are .env files present. Install python-dotenv to use them.
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
d4bc35ebe5af	day05-web	0.01%	21.21MiB / 7.498GiB	0.28%	0B / 0B	9.3MB / 147kB	1

```
^Z
[1]+  Stopped                  docker stats
```

```
mansi@MANSI: /mnt/d/INTEI x + v
GNU nano 7.2 backup.sh *
#!/bin/bash

mkdir -p backups

docker inspect day05-web > backups/container_backup.json

echo "Backup Created Successfully"
```

^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^_ Replace ^U Paste ^J Justify ^_/ Go To Line M-E Redo M-G Copy

```
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ mkdir monitoring
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ cd monitoring/
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05/monitoring$ nano backup.sh
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05/monitoring$ cd ..
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ sudo chmod +x monitoring/backup.sh
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ ./monitoring/backup.sh
Backup Created Successfully
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps/Day05$ ls backups
container_backup.json
```

```

mansi@MANSI:/mnt/d/INTERNSHIP/DevOps$ git add .
git commit -m "Day05: Added Docker Compose setup with environment variables and restart policy"
[main 0e2c93d] Day05: Added Docker Compose setup with environment variables and restart policy
3 files changed, 597 insertions(+)
create mode 100644 Day05/README.md
create mode 100644 Day05/backups/container_backup.json
create mode 100644 Day05/monitoring/backup.sh
mansi@MANSI:/mnt/d/INTERNSHIP/DevOps$ git push origin main
Username for 'https://github.com': Radhika-08
Password for 'https://Radhika-08@github.com':
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 5.17 KiB | 211.00 KiB/s, done.
Total 8 (delta 2), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Radhika-08/DevOps.git
d6a43ae..0e2c93d  main -> main

```

Workflow runs - Radhika-08/DevOps

https://github.com/Radhika-08/DevOps/actions

Radhika-08 / DevOps

Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

Actions New workflow

All workflows

Day05 Deployment Pipeline

DevOps Day04 CI

Management

Caches

Attestations

Runners

Usage metrics

Performance metrics

All workflows

Showing runs from all workflows

4 workflow runs

Workflow	Event	Status	Branch	Actor	Time	More
Day05: Added Docker Compose setup with environment vari...	main	Success	main	Radhika-08	1 minute ago	...
Day05: Added Docker Compose setup with environment vari...	main	Success	main	Radhika-08	1 minute ago	...
Added Day05 DevOps Task	main	Success	main	Radhika-08	15 minutes ago	...
Added Day05 DevOps Task	main	Success	main	Radhika-08	15 minutes ago	...
Day04 Screenshots	main	Success	main	Radhika-08	Today at 3:51 AM	...
Added Day04 DevOps Task	main	Success	main	Radhika-08	Today at 3:38 AM	...